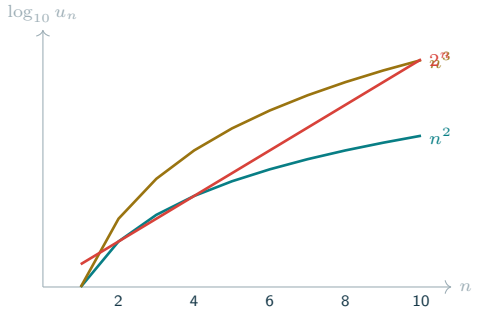
SEQUENCES LADDER 9 OF 9

The dictionary and the exceptions

Finite differences mirror calculus term by term, and sequences whose differences never settle are the exponentials.

1 / THE RULE AND THE PICTURE

Polynomials bend on a log plot and exponentials run straight



A polynomial of degree k has differences that fall to a constant, while an exponential keeps its differences proportional to itself, and on a plot of the logarithm the polynomial flattens and the exponential is a straight line:

$$\boxed{\Delta\, r^n = (r-1)\, r^n}$$

The sequence 2^n has $\Delta 2^n = 2^n$ exactly, so its differences are the sequence again.

2 / THE DICTIONARY

Each finite idea has a continuous twin

finite world	continuous world	link
difference Δ	derivative $\frac{d}{dx}$	chord gradient against tangent gradient
sum $\sum$	integral $\int$	rectangles against area
falling factorial $n^{\underline{k}}$	power x^k	$\Delta n^{\underline{k}} = k n^{\underline{k-1}}$ against $k x^{k-1}$
binomial $\binom{n}{k}$	$x^k/k!$	the basis of Newton and Taylor series
2^n, with $\Delta 2^n = 2^n$	e^x, with $\frac{d}{dx} e^x = e^x$	the function equal to its own rate of change
telescoping $\sum \Delta f = f(b) - f(a)$	$\int f' = f(b) - f(a)$	the fundamental theorem

3 / HOW TO CLASSIFY A SEQUENCE

Look at the differences and the ratios

what the differences do	what the rule is	example
constant k-th differences	polynomial of degree k	$1, 4, 9, 16$
constant ratio r	geometric, $u_n = u_1 r^{n-1}$	$3, 6, 12, 24$
Δu_n is a multiple of u_n	exponential, $u_n = c r^n$	$\Delta 3^n = 2 \cdot 3^n$
differences reproduce the sequence shifted	linear recurrence	Fibonacci: $\Delta F_n = F_{n-2}$

4 / WHERE e COMES FROM

The sequence whose difference quotient reproduces it

Take $f(x) = (1+h)^{x/h}$. Then $f(x+h) - f(x) = f(x) \cdot h$, so the difference quotient is exactly $f(x)$ for every step h, and as $h \to 0$ the function tends to e^x, the function that is its own derivative.
At $x = 1$ the values $(1 + \frac{1}{n})^n$ are $2,\ 2.5937,\ 2.7048,\ 2.7169$ for $n = 1, 10, 100, 1000$, heading to $e = 2.71828$.

5 / THE FIBONACCI NUMBERS

Differences that return the sequence

For $1, 1, 2, 3, 5, 8, 13$ the differences $0, 1, 1, 2, 3, 5$ are the same sequence moved two places along, since $F_{n+1} - F_n = F_{n-1}$, so no row of differences ever becomes constant.
Such sequences are exponential in disguise, $F_n \approx \varphi^n/\sqrt{5}$, and the characteristic equation $r^2 = r + 1$ finds the rate $\varphi = 1.618$.

6 / TRY THESE

Classify and predict

(a) For $3, 6, 12, 24$ the ratio is 2, so geometric with $u_n = 3 \cdot 2^{n-1}$. **(b)** For $1, 8, 27, 64$ the third differences are 6, so polynomial, n^3. **(c)** For $2, 5, 10, 17, 26$ the second differences are 2, so $n^2 + 1$, and for $5, 15, 45, 135$ the ratio is 3.

ASIDE / THE WHOLE LADDER

Starting from the step between neighbouring terms, the ladder reaches polynomials, sums of powers, the derivative, the integral and the exponential.

LINKS TO OTHER SHEETS

The ladder runs: 1 first differences, 2 second differences, 3 cubics, 4 Newton's formula, 5 sums of powers, 6 falling factorials, 7 the derivative, 8 the integral, and this dictionary; the functions ladder continues with the exponential.

Convention. Δ difference; r the common ratio; φ the golden ratio; $e = 2.71828\ldots$; a log plot shows $\log_{10} u_n$ against n.

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